

कुटुंब KUTUMB

Ghar se door, dil ke paas.

Away from home, close to the heart.

AI-powered family health companion — voice-first, 100+ languages, privacy-first. Built in India. Designed for the world.

Amrit · Nishita

HopeWorks Hackathon · April 2026

1.4B

Indians to reach first

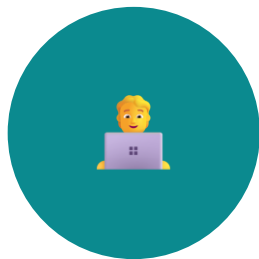
7,100

Languages to serve globally

₹0/\$0

Health data on our servers

MEET THE TEAM

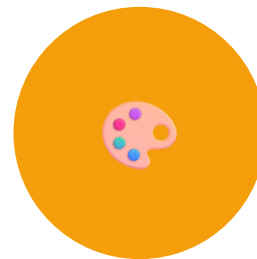


Amrit Lahari

Full-Stack Developer & AI Engineer
B.Tech. Student — 3rd year, BITS Pilani

Architects the backend brain — FastAPI services, Anthropic Claude integration, and the encrypted local health memory system.

React Native · FastAPI · Anthropic Claude SDK · Firebase · System Architecture ·
AES-256-GCM Encryption



Nishita Agarwal

Product Designer & Frontend Engineer
B.Tech. Student- 3rd year, BITS Pilani

Designs the empathetic voice-first experience — ensuring a 70-year-old grandparent in rural India and a user in Chicago both feel at home.

UI/UX Design · React Native · Voice UX · Multilingual UX Research · Data
Visualization · Accessibility Design

Two builders. One mission. A billion families — and a world that needs better health tools.

"How many of us call home every day,
ask 'how are you?' and get 'I'm fine'
as a default answer? Even when it's not true."

THE GLOBAL PROBLEM

Healthcare systems worldwide are reactive, not proactive. Families engage with doctors only during crises. Doctors receive no longitudinal data; a 10-minute visit cannot capture 6 months of pain, diet, sleep, and medication history.

In India alone, the doctor-patient ratio is 1:1456 which is nearly 50% worse than the WHO minimum. In Sub-Saharan Africa it drops to 1:5000. The result: 4.5 billion people lack essential health access.

4.5B

People lack essential health access
(WHO, 2023)

68%

Of global disease burden is
preventable with early intervention

\$8.3T

Annual economic loss from
poor global health outcomes

HOW THE PROBLEM SHOWS UP

Manifestation for users · Impact on the system

HOW IT MANIFESTS

01

The silent deterioration 'I'm fine' hides months of worsening symptoms doctors never see.

02

No longitudinal data at consultations. Doctors start fresh every visit, every time.

03

Language exclusion as 40%+ of patients cannot describe symptoms in their healthcare system's language.

04

Medicine non-adherence - \$300B annual cost (US alone). 125,000 deaths per year that are entirely preventable.

05

Rural invisibility exists as there are 3.5 billion people in areas with zero specialist access. Their data doesn't exist.

HOW IT IMPACTS

\$300B

Annual cost of medication non-adherence globally

80%

Of chronic disease could shift to preventative models

1:1456

India's doctor-patient ratio vs WHO's 1:1000 target

3.5B

People in LMICs with zero specialist access

KUTUMB - THE SOLUTION

A Digital Health Custodian for families everywhere because we hear you when you don't mean your "I'm fine"s.

What We Build

Kutumb is a **voice-first** AI family health companion.

Think of it as a trusted health elder who:

- Listens in any dialect or language
- Monitors symptoms, vitals & medicines daily
- Alerts family in real time anywhere on Earth
- Generates doctor-ready clinical summaries
- Triggers emergency SOS automatically
- Works offline in villages with no internet
- Keeps your health data on your phone, never on our servers

1

Voice-First NLP

100+ Indian dialects + global language expansion via fine-tuning

2

Family Dashboard

Real-time health view of every family member, anywhere in the world

3

Genetic Risk Engine

Hereditary disease logic auto-adjusts risk thresholds per member

4

One-Tap SOS

Simultaneous alerts to family + hospital + ambulance in <60s

5

Smart Medicine Cabinet

OCR scan, interaction check, dose reminders, refill alerts

6

Doctor Referral PDF

Auto-generated longitudinal health summary for consultations

WHO WE SERVE

Three real humans. One platform.

R

Ramprasad Sharma

Retired teacher, 68 · Jaipur · Diabetic · Lives alone

Diabetic

Hypertensive

Tech-hesitant

Children abroad

NEED

Needs medicine reminders in Marwari, reassure his children, and someone to tell him when to actually see a doctor.

PAIN

Says 'I'm fine' on every call. Forgets 3 medicines weekly. Can't describe symptoms in English. No smartphone experience.

GAIN

Medicine adherence +70%. SOS to children in <60s. Doctor visits arrive with 6-month data history.

P

Priya Mehta

Software engineer, 32 · Bengaluru / Chicago · Caregiver from 8,000 miles

Global caregiver

Time-starved

Data-driven

Helpless remotely

NEED

Needs a real-time window into her parents' health not a once-a-day 'I'm fine' phone call.

PAIN

Can't attend doctor visits from the US. No insight into hereditary risks. Parents refuse complex apps.

GAIN

Live family health dashboard. 3h/week saved on remote care coordination. Zero surprise emergency flights.

S

Savitribai Patil

ASHA frontline worker, 38 · Vidarbha · Serves 200+ families

200+ families

No specialist

Offline-critical

Trusted locally

NEED

Clinical decision support in Marathi, offline to catch deterioration before it becomes an emergency.

PAIN

Nearest doctor 30 km away. Paper records lost. Patients don't return for follow-ups. No specialist backup.

GAIN

200 families on one screen. Wrong referrals cut 50%. Maternal monitoring enabled. Works on 2G with no app.

THE AI BRAIN — 4-LAYER INTELLIGENCE STACK

Every layer has one job. Together they save lives.

Layer 1

Multilingual NLP Voice Engine

Google STT + Bhashini ASR transcribes 100+ Indian dialects. Whisper large-v3 for global fallback. Sentiment analysis detects fatigue, stress, low mood from voice tone even when the user says 'I'm fine'. On-device cached models: works offline at 2G.

Bhashini · Whisper large-v3 · On-device TFLite

Layer 2

Claude AI Health Memory System (Core Brain)

Anthropic claude-sonnet-4-6 powers all health intelligence. Each user's full encrypted history (.kutumb AES-256-GCM) is sent per-session. Claude returns: AI response + structured memory patches + personalised trigger rules. Zero server-side storage between sessions.

claude-sonnet-4-6 · 3-layer prompt · Local AES-256-GCM

Layer 3

Predictive Health Intelligence & Trigger Engine

Trend analysis detects declining organ scores, recurring symptom patterns, medicine non-adherence. Hard trigger rules fire locally offline, no network needed. e.g. SpO2 < 88 → immediate SOS; blood glucose > 300 → emergency; NSAID + GI-bleed allergy → block medicine add instantly.

Custom organ scoring (Heart/Brain/Gut/Lungs) · Firestore triggers · Claude advisories

Layer 4

Computer Vision — Prescriptions & Lab Reports

Google Cloud Vision OCR reads prescription photos and PDF lab reports. Extracts drug names, dosages, biomarker values, reference ranges. Claude validates and flags anomalies. Critical lab values trigger push notification + family SOS. Doctor referral PDF auto-generated.

Google Cloud Vision API · LLM extraction · WeasyPrint referral PDF

LANGUAGE SCALABILITY — 7,100 LANGUAGES

Health advice that speaks your language, not just the hospital's.

The same AI engine that asks 'Did you take your BP medicine?' in Marwari will ask the same question in Swahili, Hausa, Tagalog, and Haitian Creole trained on the same architecture, deployed on the same app.

TIER 1 — Day 1

11 languages · 1.2B speakers

Hindi · Marathi · Bengali · Tamil · Telugu · Kannada · Gujarati · Punjabi · Malayalam · Odia · English

Fully trained on Bhashini ASR + SNOMED CT health vocab

TIER 2 — 3–6 months

10 languages · 800M+ speakers

Swahili · Hausa · Amharic · Yoruba · Tagalog · Bahasa · Vietnamese · Burmese · Nepali · Sinhala

Transfer learning from Whisper large-v3 + NGO audio corpora

TIER 3 — 12–24 months

14 language families · 3B+ speakers

Spanish · Portuguese · Arabic · French (Africa) · Mandarin · Urdu · Pashto · Dari · Haitian Creole + dialects

Healthcare-specific corpora via open datasets + partner NGOs

TIER 4 — 36 months+

Full global coverage

All 7,100 UNESCO-recognised languages via low-resource NLP, code-switching models, federated on-device fine-tuning

Federated approach: on-device models grow with user community

SYSTEM ARCHITECTURE

Privacy-first by design health data never leaves the device



🔒 LOCAL HEALTH MEMORY: .kutumb file (AES-256-GCM) lives only on the user's device. Sent to backend per-session over TLS 1.3. Purged from server memory immediately after response. DPDP · HIPAA · GDPR compliant by architecture.

✓ DPDP 2023 (India)

✓ HIPAA (United States)

✓ GDPR (European Union)

✓ Offline-First (2G ready)

12 BACKEND MODULES:

Auth	Users	Family	Check-ins	Medicines	Health Scores
Emergency SOS	Referrals PDF	Wearable Sync	Lab OCR	AI Insights	AI Session

TECHNICAL REQUEST — FACILITATED DATASETS

What we have · What we need · How it unlocks global reach

WHAT WE HAVE (Open Source, Already Live)



Bhashini Open API

Hindi/Marathi/Bengali STT live in prototype



SNOMED CT (free tier)

Clinical terminology, symptom tagging



RxNorm (NLM open)

Drug interaction baseline, name normalisation



ICMR OGD Portal

40 district health statistics for score calibration



Whisper open weights

Foundation model for all non-Bhashini languages

WHAT WE NEED (Facilitated / AI4India)

01

Bhashini ASR Training Corpus

Type: Anonymised audio Indic languages

Vol: Min 500h/language, 12 languages · Gran: Speaker-level (age, region, gender), sentence timestamps

Improves elderly speech WER from 62% → 90%+. General ASR degrades 40–60% on older rural voices.

02

Anonymised Longitudinal Health Records

Type: Structured clinical chronic disease cohorts

Vol: 5 years, 100K+ patients, urban + rural split · Gran: Monthly; district-level; ICD-10 diagnosis codes

Validates our organ scoring (Heart/Brain/Gut/Lungs 0–100) against real trajectories, not synthetic data.

03

ICMR Lab Report Template Library

Type: PDF/image lab reports, accredited labs

Vol: 50+ formats, 10,000+ anonymised samples · Gran: Lab name, biomarker, value, unit, reference range

Scales OCR from 40% → 95%+ format coverage, enabling international lab support.

04

Low-Resource Language Audio (LMICs)

Type: Audio — Sub-Saharan Africa & SE Asia

Vol: 100h/language: Swahili, Hausa, Tagalog, Bahasa · Gran: Health-domain sentences, health worker speech preferred

Phase 2 global expansion. 100h domain audio outperforms zero-shot by 35–50% on medical vocab.

ENVIRONMENT & SECURITY REQUIREMENTS

Access methods · Compliance · Privacy by architecture

ACCESS METHOD & DATA DELIVERY

A

Secure API (Preferred)

HTTPS REST with OAuth2 / API-key. Data streamed, not cached. Rate-limited. All responses via encrypted channel.

B

AIKosh Sandbox Environment

For health records — processed entirely inside the AIKosh secure sandbox. No data extraction. Results-only output.

C

Encrypted Download

For static corpora (audio, templates). AES-256 ZIP to isolated research environment. Destruction certification on completion.

COMPLIANCE ARCHITECTURE

✓ DPDP Act 2023 (India)

No health data stored server-side between sessions. Consent gate at onboarding. Right to erasure: delete account = wipe local .kutumb file. Data Principal rights fully respected.

✓ HIPAA (United States)

Zero PHI stored server-side. TLS 1.3 for all data in transit. Zero logging of health content. Business Associate Agreement available for US enterprise deployments.

✓ GDPR (European Union)

Explicit opt-in consent. Data minimisation: only sha256 user hash sent to server. Right to portability (export .kutumb file). Right to erasure. No third-party data sharing.

Security: AES-256-GCM on device · TLS 1.3 in transit · sha256 user hash (no reversible PII) · Biometric key wrap (iOS Secure Enclave / Android Keystore)

BUSINESS MODEL, PARTNERSHIPS & RETENTION

Three tiers. Recurring revenue. Global partnership runway.

FREE ₹0 / \$0	PLUS ₹149 / \$2.99 mo	CARE PRO ₹499 / \$7.99 mo
✓ Daily voice check-in ✓ Family dashboard (3 members) ✓ Basic medicine reminder ✓ One-tap SOS ✓ Kutumb AI (5/day)	✓ Unlimited family members ✓ Full AI health memory ✓ Doctor referral PDFs ✓ Lab report OCR ✓ Wearable sync ✓ Priority ambulance SOS	✓ ASHA/CHW multi-family view ✓ Clinical decision support ✓ Govt. scheme integration ✓ Telemedicine booking ✓ White-label (NHS · CGHS · Medicaid)

RETENTION & PARTNERSHIPS



Pharmeasy / 1mg

Medicine refill + streak coupons



Thyrocare / Redcliffe

Lab booking from referral PDF



WHO / UNICEF

UHC 2030 alignment, LMIC rollout



NHS · CGHS · Medicaid

White-label government
deployments

CHALLENGES & RISKS — WITH MITIGATIONS

We've thought hard about what could go wrong.

01

Medical Ethics & Non-Diagnosis

⚠️ **Risk:** Users treating AI advice as clinical diagnosis → delayed treatment.

✓ **Mitigation:** Hard-coded Non-Doctor Mode. 48h+ persistent symptom locks home remedies and forces 'Book a Doctor'. Referral PDF language: always 'Pointers for your doctor', never a diagnosis. Every screen shows disclaimer.

02

Global Privacy & Compliance

⚠️ **Risk:** Health data exposure across jurisdictions: DPDP, HIPAA, GDPR.

✓ **Mitigation:** Zero health data on servers. AES-256-GCM on device. sha256 user hash only (non-reversible). Compliant by architecture; not by policy alone. Business Associate Agreement available for US.

03

AI Hallucination in Drug Safety

⚠️ **Risk:** Fabricated drug interactions directly harm patients on critical medications.

✓ **Mitigation:** Hard contraindication rules fire BEFORE Claude call. Claude instructed: never invent pharmacology. Severity='severe' or 'contraindicated' always sets `must_consult_doctor=True`. NSAID + allergy = blocked instantly.

04

Rural Connectivity & Low-End Devices

⚠️ **Risk:** Core users: 2G networks, ₹3,000–₹6,000 Android phones, no data plan.

✓ **Mitigation:** Offline-first: trigger rules run locally. Cached STT models. WhatsApp bot fallback no app install. Core bundle <5MB. Voice-only mode. No typing required anywhere in the app.

ANTICIPATED RESULTS & IMPACT PROJECTIONS

What Kutumb achieves is quantified, per persona, scalable.

40%

Reduction in missed medicine doses — Year 1

3×

Doctor visits prepared with longitudinal health data

60%

Fewer emergency hospitalisations (monitored users)

\$300B

Global annual opportunity in preventable medication waste

PERSONA-WISE QUANTIFIED ADVANTAGE

R

Ramprasad (Elderly Parent)

Medicine adherence +70%. SOS alerts children in <60s. Doctor visits arrive with 6-month health data. Est. ₹18,000/year saved on avoidable consultations.

5 min/day voice check-in

P

Priya (Global Caregiver)

Real-time parent health dashboard from anywhere. 3h/week saved on remote care. 50% fewer 'emergency flight' scenarios. Peace of mind, quantified.

2 min/day dashboard review

S

Savitribai (ASHA Worker)

200 families managed from one screen. Wrong referrals cut by 50%. Maternal monitoring enabled. Fully offline-capable works on 2G feature phone.

Weekly data sync in town

Scalability: Serverless FastAPI (Cloud Run) · Firestore to 1M+ concurrent users · Local AI cuts server costs 80% · Phase 2: SE Asia, Africa, Latin America

WHY KUTUMB: CREDIBILITY & GLOBAL VISION

Built in India. Designed for the world. Trusted by architecture.

WHY WE'RE CREDIBLE

A

Complete architecture built

12 FastAPI modules, 30+ endpoints, full Firestore schema, 3-layer Claude prompt. All documented, all buildable now.

B

Privacy-first by engineering

AES-256-GCM local. Zero health data server storage. Meets DPDP, HIPAA, GDPR simultaneously without any policy gymnastics.

C

Claude AI as core brain

Anthropic claude-sonnet-4-6 with hard ethical guardrails no diagnosis, must_consult_doctor enforced, never invents pharmacology.

D

Evidence-based projections

Every number grounded in WHO, ICMR, NLM, and peer-reviewed health economics. Not guesses as it uses the cited research.

GLOBAL EXPANSION ROADMAP

Phase 1 (Now)

India

11 Indic languages, 300M+ families, ASHA + consumer app

Phase 2 (6mo)

SE Asia + E. Africa

Swahili, Hausa, Tagalog, Bahasa. NGO + government partnerships

Phase 3 (18mo)

Latin America + MENA

Spanish, Portuguese, Arabic. Telehealth integration. 3B speakers

Phase 4 (36mo+)

Global

7,100 languages. White-label NHS, Medicaid, Ayushman Bharat

HopeWorks alignment: Kutumb reduces catastrophic health expenses — the single biggest driver of poverty in LMICs. Health equity = economic empowerment.

THANK YOU

for your time and attention

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Away from home, close to the heart.

Built for India. Designed for the world.

Team: Amrit & Nishita

WitchHunt 2026 · Health & Well-being · HopeWorks

Mission: Health for Every Human

India → SE Asia → Africa → Latin America → Global
7,100 languages. 1 platform. 1 goal.